

FINAL PROJECT 2024/2025

Queueing Theory Simulation Report: McDonald Mulyosari Self-Service Machine

Anggota:

Muhammad Arham Dwi Nugroho	(5003231001)
Abid Humamrusydan	(5003231045)
Razaki Andyatama Ken Onodera	(5003231098)

Dosen Pembimbing :

Diaz Fitra Aksioma, S.Si., M.Si.

**PROGRAM STUDI SARJANA STATISTIKA DEPARTEMEN
STATISTIKA**

**FAKULTAS SAINS DAN ANALITIKA DATA INSTITUT
TEKNOLOGI SEPULUH NOPEMBER SURABAYA**

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Members:

Muhammad Arham Dwi Nugroho	(5003231001)
Abid Humamrusydan	(5003231045)
Razaki Andyatama Ken Onodera	(5003231098)

Supervising Lecturer:

Diaz Fitra Aksioma, S.Si., M.Si.

**BACHELOR OF STATISTICS STUDY PROGRAM
DEPARTMENT OF STATISTICS
FACULTY OF SCIENCE AND DATA ANALYTICS
SEPULUH NOPEMBER INSTITUTE OF TECHNOLOGY
SURABAYA
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INTRODUCTION

1.1 Background of Study

In recent years, the integration of digital technology into the food service industry has significantly changed how customers interact with fast food restaurants. As part of this trend, McDonald's has adopted self-order machines, or self-service kiosks, to enhance service speed, reduce reliance on front-line staff, and improve the overall ordering experience.

These machines are designed to streamline the ordering and payment process by allowing customers to place and pay for their orders directly through a digital interface. In theory, this leads to reduced wait times, increased order accuracy, and higher throughput during peak hours. However, the actual efficiency of self-order systems may vary based on store layout, user familiarity with the interface, and traffic conditions.

To evaluate the operational performance of such a system, it is necessary to study how effectively it handles customer flow. One widely used approach for this is queueing theory, which models and analyzes the behavior of waiting lines. By applying queueing models, such as M/M/1, researchers can measure key performance metrics like arrival rate, service rate, waiting time, and server utilization.

This study applies the M/M/1 queueing model to assess the performance of two independent self-service counters at McDonald's Mulyosari branch in Surabaya. Observations were carried out during weekends and national holidays surrounding the *Idul Adha* period, a time when customer volume was relatively low due to the cultural tradition of *pulang kampung* (returning to hometowns). As a result, queue buildup was minimal during the observation period. Nonetheless, this scenario provided a unique opportunity to evaluate baseline system performance under light traffic conditions using real-world data.

1.2 Research Problems

As fast-food chains increasingly adopt self-service kiosks to improve operational efficiency, questions remain about how effectively these systems manage customer flow under real-world conditions. While self-order machines are designed to reduce wait times and streamline service, their actual performance in queue management is not always clear.

In high-traffic locations like McDonald's, inefficient queue management can lead to service delays and reduced customer satisfaction. Understanding the performance of self-service counters through quantitative modeling is essential for identifying system bottlenecks and assessing capacity.

This study addresses the problem by applying the M/M/1 queueing model to two independent self-order counters at McDonald's Mulyosari. The goal is to evaluate whether the current system effectively handles customer arrivals without significant buildup, even during observed periods like weekends and national holidays.

1.3 Objective of study

The main objective of this study is to evaluate the operational performance of McDonald's self-service ordering system using queueing theory, specifically the M/M/1 model. The study focuses on two independent self-service counters at McDonald's Mulyosari branch and aims to analyze system performance under actual customer flow conditions. This is achieved by collecting real-time observational data and calculating key performance metrics, including:

- Arrival rate (λ)
- Service rate (μ)
- Utilization (ρ)
- Average waiting time (W_q)
- Average time in system (W)
- Expected queue length (L_q)
- Expected number in system (L)

The goal is to determine whether the current system can efficiently handle customer demand during operational hours, particularly during weekends and holidays, when unique traffic patterns occur. The results provide a quantitative basis for assessing the effectiveness of self-order kiosks in managing service flow.

1.4 Benefits of study

This study contributes to both practical and academic understanding of operational efficiency in quick-service restaurant settings by applying queueing theory to evaluate self-service ordering systems. Using real-time observational data and performance modeling, the study offers quantitative insights into how self-service counters perform under real-world traffic conditions.

For practitioners and store managers, the findings provide a data-driven basis for assessing whether the current configuration of self-service counters is sufficient to meet customer demand. Metrics such as arrival rate, service rate, and utilization can help identify underused capacity, validate current resource allocation, or signal potential inefficiencies under different traffic scenarios.

From an academic perspective, this research demonstrates the application of the M/M/1 queueing model in analyzing service systems using real data. It adds to the operational research literature by showcasing how theoretical models can be used to assess live service environments like fast-food restaurants.

By focusing on measurable outcomes—such as waiting time, queue length, and system utilization—this study supports more objective evaluations of self-service technology and helps guide future data-informed improvements in service design.

1.5 Scope and Limitations

Scope

This study focuses on analyzing the queueing performance of the self-service ordering system at the McDonald's Mulyosari branch in Surabaya. Specifically, the research applies the M/M/1 queueing model to each of the two independent self-service counters, based on observational data collected during both peak and off-peak hours. The scope of the study includes:

- Recording timestamps of customer arrival, system start (order begin), system end (payment complete), and pickup time
- Calculating key performance metrics:
Arrival rate (λ), Service rate (μ), Utilization (ρ), Mean waiting time (W_q), Mean time in system (W), Expected queue length (L_q), and Expected number in system (L)
- Evaluating whether the observed interarrival and service times are consistent with the exponential distribution assumption required by the M/M/1 model

Limitations

- Distribution Assumptions: The M/M/1 model assumes exponential interarrival and service times. Real-world data may only approximately follow these distributions.
- Independent Queues: Each self-service counter is modeled as a separate queue. Customer switching behavior or shared queue dynamics are not considered.
- Limited Observation Window: Data was collected over a short period during weekends and national holidays, potentially limiting generalizability to regular weekdays or long-term trends.
- Pickup Excluded from Service Time: Although pickup time was recorded, it is not included in the service duration used for queueing calculations.
- Single Location: The study is restricted to one McDonald's outlet, and results may not be representative of other locations with different layouts or customer volumes.

LITERATURE REVIEW

2.1 Queueing System

A queueing system is a mathematical model used to analyze situations where customers or items wait for service. It provides a framework to study important system characteristics such as customer waiting time, server utilization, and the number of customers present in the system. Queueing theory is widely applied in various service industries, including restaurants, banks, hospitals, manufacturing systems, and call centers, where managing customer flow and service efficiency is essential.

2.1.1 Types of Queueing System

There are various types of queueing systems, which differ based on how customers arrive, how services are provided, and how many servers are involved. In this study, the focus is on the M/M/1 model (Arrival / Service / Server), which can be described as follows:

- **M (Markovian Arrival):** Customer arrivals follow a Poisson process, meaning arrivals occur randomly and independently over time.
- **M (Markovian Service):** Service times follow an exponential distribution, indicating that the time to serve each customer is random and memoryless.
- **1 (Single Server):** The system has only one server or service counter.

Since the McDonald's Mulyosari branch operates two independent self-service counters, each counter is modeled as a separate M/M/1 queueing system.

2.1.2 Measures of Performance

Several important performance measures are used in queueing theory:

- **Arrival Rate (λ):** Expected number of customers arriving per time unit.
- **Service Time (S):** Expected time taken to serve one customer.
- **Service Rate (μ):** The number of customers that can be served per time unit $\mu=1/S$.
- **Utilization (ρ):** The proportion of time the server is busy, calculated by $\rho=\lambda/\mu$.
- **Mean Waiting Time in Queue (Wq):** Expected time spent waiting before being served.
- **Mean Time in System (W):** The total time customers spend in the system (waiting + service).
- **Average Number in Queue (Lq):** Expected number of customers waiting.
- **Average Number in System (L):** Expected number in the system.

All formulas assume a steady-state system with $\rho < 1$.

2.2 Steady State

The steady-state condition refers to a system where queue performance becomes stable over time. This means that customer arrivals and service completions occur at consistent average rates, and performance measures like waiting time and queue length do not grow uncontrollably. For the M/M/1 model, the system reaches steady state if:

$$\rho = \frac{\lambda}{\mu} < 1$$

where:

λ = average arrival rate (customers per time unit),

μ = average service rate (customers served per time unit),

ρ = system utilization.

This ensures that the service facility can handle customer demand in the long run. If $\rho \geq 1$, the system is unstable, meaning the queue will grow indefinitely.

2.3 Poisson Process and Exponential Distribution

In queueing theory, customer arrivals are commonly modeled as a Poisson process, which assumes that arrivals occur randomly and independently over time at a constant average rate. This implies that the number of arrivals within a fixed time interval follows a Poisson distribution.

As a direct result, the time between consecutive arrivals—known as the interarrival time—follows an Exponential distribution. This distribution describes the probability that a certain amount of time will pass before the next arrival occurs and is characterized by a single parameter: the average arrival rate (λ).

Similarly, in the M/M/1 queueing model, the service times are also assumed to follow an Exponential distribution, reflecting the assumption that service durations are random, memoryless, and identically distributed across customers, with a constant average service rate (μ).

A key mathematical property of the Exponential distribution is it is memoryless, meaning that the probability of an event (such as the next arrival or service completion) occurring in the future does not depend on how much time has already passed. This property simplifies the analysis of queueing systems and enables the derivation of closed-form performance metrics in M/M/1 models.

2.4 McDonald

The self-service ordering system at McDonald's Mulyosari branch in Surabaya presents a practical real-world application for queueing theory. The branch is equipped with two independent self-service kiosks, where customers place and complete payment for their orders without human cashier interaction. Each kiosk operates independently and serves its own line of customers, forming two separate queueing systems.

This setup is well-suited for analysis using the M/M/1 queueing model, which assumes a single server, Poisson arrivals, and exponentially distributed service times. The independent operation of the kiosks ensures that customer arrivals and service processes at each kiosk can be treated separately under this model. Each kiosk also follows a First-Come, First-Served (FCFS) discipline, with no shared queue or server coordination.

After completing their order, customers proceed to a shared pickup counter to collect their food. However, this pickup stage is not included in the modeled service time for this study, as the focus remains solely on the ordering and payment process conducted through the kiosks.

Additionally, this study only includes cashless transactions completed through the self-service kiosks. Cash-paying customers and those ordering at traditional counters were excluded to ensure consistency in the modeled service structure.

Research Methodology

3.1 Data Source

The primary data for this study was obtained through direct observation at the McDonald's Mulyosari branch in Surabaya. Observations were focused on the self-service ordering system, which includes two independent self-order kiosks.

Only cashless transactions completed through the kiosks were included to ensure consistency in the ordering and payment process. Customers paying at traditional counters or using alternative methods were excluded from the study. For each customer, the following timestamps were recorded:

- Arrival Time — when the customer approached the kiosk or joined the line
- Order Start Time — when the customer began interacting with the screen
- Order End Time — when the payment process was completed
- Pickup Time — when the customer received their food from the pickup counter
- Counter ID — identifying which of the two kiosks was used

From these observations, key metrics such as service time, waiting time, interarrival time, and total time in system were later calculated. Data collection was conducted unobtrusively to minimize behavioral bias (Hawthorne effect), using synchronized digital clocks to ensure timestamp accuracy.

In addition to primary data, relevant secondary sources, including academic literature on queueing theory and industry studies on self-service technologies, were reviewed to support the analysis and provide theoretical grounding for the M/M/1 modeling approach.

3.2 Research Variables

The variables used in this study are organized into input, process, and output categories, reflecting the flow of customers through the self-ordering system. These variables were either recorded directly through observation or calculated from the observed timestamps.

Input Variables:

- Arrival Time: The time when a customer arrived at the self-service area.
- Interarrival Time: The time difference between the arrivals of successive customers; calculated from the arrival timestamps.

Process Variables:

- Order Start Time: The time when a customer began interacting with the kiosk.
- Order End Time: The time when order and payment were completed.
- Service Time: The duration of the customer's interaction with the kiosk; calculated as Order End Time – Order Start Time.
- Waiting Time: The time the customer waited before starting service; calculated as Order Start Time – Arrival Time.

Output Variables

- Total Time in System: The sum of waiting time and service time; calculated as Order End Time – Arrival Time.
- Pickup Time: The time when the customer received their order at the collection counter.
- Pickup Delay: The time between order completion and food pickup; calculated as Pickup Time – Order End Time. Note: This is recorded but not included in the M/M/1 model.

These variables were used to derive key performance metrics for each self-service counter, including:

- Arrival Rate (λ)
- Service Rate (μ)
- Utilization (ρ)
- Mean Waiting Time in Queue (W_q)
- Mean Time in System (W)
- Expected Number in Queue (L_q)
- Expected Number in System (L)

These model-based performance measures were computed using standard formulas for the M/M/1 queueing model under steady-state assumptions.

3.3 Data Structure

The dataset consists of time-stamped records collected through direct observation of customer interactions with the self-order kiosks at McDonald's Mulyosari branch. Each row in the dataset corresponds to a single transaction and includes the following raw data fields:

- Counter ID (1 or 2): Indicates which self-service kiosk the customer used
- Arrival Time: When the customer arrived at the self-service area
- System Start Time: When the customer began interacting with the kiosk
- System End Time: When the customer completed their order and payment
- Pickup Time: When the customer received their food at the pickup counter

From these observations, several derived variables were computed:

- Waiting Time = System Start Time – Arrival Time
- Service Time = System End Time – System Start Time
- Total Time in System = System End Time – Arrival Time
- Pickup Delay = Pickup Time – System End Time
- Interarrival Time = Arrival[n] – Arrival[n-1] (calculated separately for each counter)

This structured dataset allows for the application of M/M/1 queueing formulas to each counter independently. The derived time intervals support the estimation of performance metrics such as arrival rate (λ), service rate (μ), waiting time (W_q), and system utilization (ρ).

3.4 Service System Layout

The service system at McDonald's Mulyosari branch is composed of two main stages:

Ordering and Payment Stage: Customers place and pay for their orders at one of two independent self-service kiosks (designated as Counter 1 and Counter 2). Each kiosk operates as a separate, single-server queue without a shared line or central dispatcher. Customers approach either kiosk freely, forming two parallel and independent queueing systems.

Food Pickup Stage: After completing the ordering process, all customers proceed to a shared pickup counter, where their food is prepared and delivered. This stage involves waiting for food preparation but is not included in the queueing model.

For the purpose of analysis, only the ordering and payment process at the kiosks is modeled. Each kiosk is treated as an independent M/M/1 queue, where arrival rate (λ), service rate (μ), and performance measures are calculated separately based on observed data.

This layout supports the assumption of two parallel, memoryless systems with exponential arrival and service patterns, suitable for analysis using steady-state queueing theory.

3.5 Service Stages

The customer service process at McDonald's Mulyosari self-service kiosks is divided into several stages that reflect the flow of interaction from arrival to food collection. These stages are described as follows:

- Queueing Stage: The customer arrives at the self-order area. If both kiosks are currently in use, the customer waits until one becomes available. The time from arrival to the start of interaction is recorded as waiting time.
- Ordering and Payment Stage: Once a kiosk is available, the customer uses the touchscreen interface to select items, customize the order, and complete payment. The duration from the beginning to the end of this interaction is defined as service time.
- Post-Service Transition Stage: After completing the order and payment, the customer receives a receipt and proceeds to the pickup counter. This stage is observed but not included in the queueing model.
- Food Pickup Stage: At the shared pickup counter, the customer waits for their order number to be called and collects the food. The time between the end of the kiosk interaction and the food pickup is recorded as pickup delay. This stage is also excluded from the M/M/1 model, as it does not involve the kiosk service system.

3.6 Modeling Assumptions

To apply the M/M/1 queueing model effectively, the following assumptions are made in this study:

- Poisson Arrivals: Customers arrive according to a Poisson process, meaning the time between arrivals is exponentially distributed.
- Exponential Service Times: The duration to complete both order and payment follows an exponential distribution.

- Single Server per Queue: Each self-service counter is treated as an independent single-server queue.
- First Come, First Served (FCFS): Customers are served in the order they arrive.
- Unlimited Queue Capacity: There is no upper limit on the number of customers waiting in line.
- Independent Processes: Arrival and service times are independent from each other and across counters.
- Pickup Excluded: The food pickup stage is not considered part of the queueing service time.

RESULT AND DISCUSSION

4.1 Data Characteristics

The data for this study was collected through direct observation at McDonald's Mulyosari branch in Surabaya. The observation focused on the self-service ordering system, which consists of two independent self-service counters. Each self-service counter was treated as a separate queueing system.

The data collection took place during weekends and national holidays, specifically around the *Idul Adha* period, when many individuals participate in the tradition of *pulang kampung* (returning to hometowns). This timing resulted in relatively low customer traffic, and no significant queue buildup was observed throughout the observation period. Despite the low-demand conditions, the analysis was done for evaluating system performance and establishing baseline metrics under real-world conditions. For each observed customer, the following timestamps were recorded:

- Counter (Self-service machine used)
- Arrival Time (time customer arrived at the queue)
- System Start Time (time customer started using the self-service machine)
- System End Time (time customer completed ordering and payment)
- Pickup Time (time customer picked up their food at the collection counter)

From these primary timestamps, additional variables were calculated for analysis:

- Total Time in System = System End Time – Arrival Time
- Waiting Time = System Start Time – Arrival Time
- Service Time = System End Time – System Start Time
- Pickup Delay = Pickup Time – System End Time
- Interarrival Time = difference between successive Arrival Times
- Time Between System Starts = difference between successive System Start Times

The data collection was conducted across multiple time periods. This dataset serves as the basis for parameter estimation, queueing model calculation, and performance analysis in the following sections. The total number of observations collected was 71.

4.2 Parameter Estimation

Based on the observed data, the arrival rate and service rate parameters for each self-service counter were estimated as follows:

Arrival Rate (λ): The arrival rate represents the average number of customers arriving per unit time. It was calculated as:

$$\text{Arrival Rate } (\lambda) = \frac{1}{\text{Average (Time between arrival)}}$$

Time between arrival = interarrival time

Service Time (S): The service time represents the average time required for a customer to complete the ordering and payment process at the self-service kiosk.

$$\text{Service Time (S)} = \text{System end} - \text{System Start}$$

System End = the time when the customers finished ordering and paying

System Start = the time when the customer began using the kiosk

Service Rate (μ): The service rate represents the number of customers that can be served per unit time, calculated as:

$$\text{Service Rate } (\mu) = \frac{1}{\text{Service Time (S)}}$$

The estimated parameters for each server are summarized as follows:

Metric	Server 1	Server 2	Unit
Arrival Rate (λ)	7.58	6.17	customers/hour
Service Time (S)	0:02:38	0:02:49	HH:MM:SS
Service Rate (μ)	22.80	21.36	customers/hour

These parameter estimates serve as the input for the queueing model calculations in the next section.

To visually assess whether the assumptions of exponential service and interarrival times are reasonable, histograms were generated for each metric. The distribution of interarrival times appeared right-skewed, with many short intervals and fewer long ones—consistent with the Poisson arrival process assumed by the M/M/1 model. Similarly, the service time distribution was also skewed, supporting the exponential service time assumption. The histograms are presented in **Appendix B (Figures B.1–B.4)** as visual justification for the theoretical model used.

4.3 Queueing Model Calculations

Using the estimated arrival rate (λ) and service rate (μ) from the previous section, the performance measures of the queueing system were calculated based on the standard formulas for the M/M/1 model:

Utilization (ρ):

$$\rho = \frac{\lambda}{\mu}$$

This represents the proportion of time the server is occupied.

Probability of Zero Customers (P0)

$$P_0 = 1 - \rho$$

This indicates the probability that the system is idle.

Average Waiting Time in Queue (W_q):

$$W_q = \frac{L_q}{\lambda} = \frac{\rho}{\mu(1 - \rho)}$$

Average Time in System (W):

$$W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}$$

Average Number of Customers in Queue (L_q):

$$L_q = \frac{\rho^2}{1 - \rho}$$

Average Number of Customers in System (L):

$$L = \frac{\rho}{1 - \rho}$$

The calculated performance measures for both self-service counters are summarized in the following table:

Metric	Server 1	Server 2	Unit
Utilization (ρ)	0.33	0.29	proportion
P_0	0.67	0.71	probability
W_q	0.02	0.02	hours
W	0.07	0.07	hours
L_q	0.17	0.12	customers
L	0.50	0.41	customers

These results describe the expected performance of each queueing system under steady-state conditions.

4.4 System Performance Discussion

The queueing performance metrics calculated in the previous section provide insights into the operational efficiency of the self-service kiosks at McDonald's Mulyosari branch. Both counters exhibited low utilization rates ($\rho = 0.33$ for Counter 1 and $\rho = 0.29$ for Counter 2), indicating that the kiosks were not operating at full capacity during the observation period. This is consistent with the context of data collection, which took place during weekends and national holidays surrounding the Idul Adha period, when customer traffic was relatively low.

The probability that a kiosk was idle (P_0) was high for both counters (0.67 and 0.71), suggesting that customers were rarely required to wait in line. This is further

supported by the average waiting times in queue (W_q) of only 0.02 hours (approximately 1.2 minutes) and average total time in the system (W) of around 0.07 hours (approximately 4.2 minutes). These values reflect a smooth and efficient service process under low-load conditions.

Furthermore, the histograms presented in Appendix B illustrate the skewed distribution of both interarrival and service times, providing visual support for the exponential distribution assumption used in the M/M/1 model. The short waiting times and minimal queue buildup observed during the study reinforce the model's validity for this setting.

Overall, the results suggest that the current self-service system at McDonald's Mulyosari is capable of handling the observed traffic efficiently. However, it is important to note that the findings reflect off-peak performance and may not generalize to high-demand periods. Future evaluation under heavier traffic conditions is recommended to assess the system's performance limits and potential for congestion.

Conclusion

5.1 Conclusion

This study evaluated the performance of self-service ordering kiosks at the McDonald's Mulyosari branch using the M/M/1 queueing model. Observational data were collected from two independent kiosks during weekends and national holidays surrounding the Idul Adha period, a time of relatively low customer volume. A total of 71 valid observations were recorded, including timestamps for arrival, service initiation, service completion, and pickup.

The queueing analysis focused on estimating key performance metrics, including arrival rate (λ), service rate (μ), server utilization (ρ), average waiting time (W_q), and average number of customers in the system (L). The estimated arrival rates were 7.58 and 6.17 customers per hour for Server 1 and Server 2, respectively, while the corresponding service rates were 22.80 and 21.36 customers per hour. These values resulted in relatively low server utilizations of 0.33 and 0.29, indicating that both servers were under capacity and capable of handling higher traffic without congestion.

The calculated performance measures showed minimal queue buildup, with average waiting times (W_q) of only 0.02 hours (approximately 1.2 minutes) and average total time in the system (W) of about 0.07 hours (approximately 4.2 minutes) for both servers. These results are consistent with the low customer volume observed during the data collection period and demonstrate that the current configuration of self-service kiosks is effective for handling light traffic conditions.

Overall, the study confirms that under low-demand conditions, the self-service kiosks at McDonald's Mulyosari perform efficiently with minimal waiting times and low system congestion. The M/M/1 queueing model proved appropriate for analyzing the system and establishing baseline performance metrics.

5.2 Suggestion

Based on the findings of this study, several recommendations can be made for both future research and operational improvements:

Extended Observation Period: Future studies should include data collection during regular weekdays and peak meal times to evaluate kiosk performance under high-traffic conditions. This would allow for a more comprehensive understanding of system capacity and bottlenecks.

Inclusion of Pickup Delay in Modeling: While the pickup stage was excluded from the queueing model, it may affect overall customer experience. Incorporating pickup delay

into a multi-stage or network queueing model could provide a more complete performance evaluation.

Comparative Analysis with Staffed Counters: Comparing the performance of self-service kiosks with traditional cashier counters could offer insights into which system is more efficient under different customer load scenarios.

Interface and User Flow Study: While this study focused on timing metrics, future research could explore how kiosk interface design and customer familiarity influence service time variability.

Real-Time Monitoring System: Implementing a basic tracking system to monitor kiosk usage and queue lengths could help store managers make informed decisions about staff deployment and kiosk placement.

These suggestions aim to support both continued academic inquiry and practical operational enhancements in fast-food queue management using self-service technology.

Reference

- Hillier, F. S., & Lieberman, G. J. (2001). *Introduction to Operations Research* (7th ed.). McGraw-Hill.
- Diaz, F. Aksioma. (n.d.). *Queueing Theory (ITS Lecture Slides)*. Institut Teknologi Sepuluh Nopember.

Appendices

Appendix A

Cleaned Data

Link to table: https://docs.google.com/spreadsheets/d/1LwVg3Lg8hT-V22s_NXZbMd4HcRrA55aID2uXGhNcrDk/edit?usp=sharing

No	Subject	Counter	Arrival Time	System Start	System End	Pickup	Total Time System	Waiting Time	Pickup Delay	InterArrival Time	Time Between System Start
1	baju ijo orang 40 an	1	13:30:38	13:33:14	13:34:58	13:40:17	0:04:20	0:02:36	0:05:19	#VALUE!	#VALUE!
2	Girl, White clothes, chocolate hijab	1	13:38:24	13:38:24	13:40:23	13:48:14	0:01:59	0:00:00	0:07:51	0:07:46	0:05:10
3	Man, baju ijo, 40 ish	1	13:40:55	13:40:55	13:44:16	13:45:28	0:03:21	0:00:00	0:01:12	0:02:31	0:02:31
4	Ibu bayii	1	13:44:19	13:44:19	13:45:34	13:52:36	0:01:15	0:00:00	0:07:02	0:03:24	0:03:24
5	Mbak jacket Putih	1	13:51:17	13:51:17	13:52:23	13:58:46	0:01:06	0:00:00	0:06:23	0:06:58	0:06:58

Appendix B – Histogram Plots

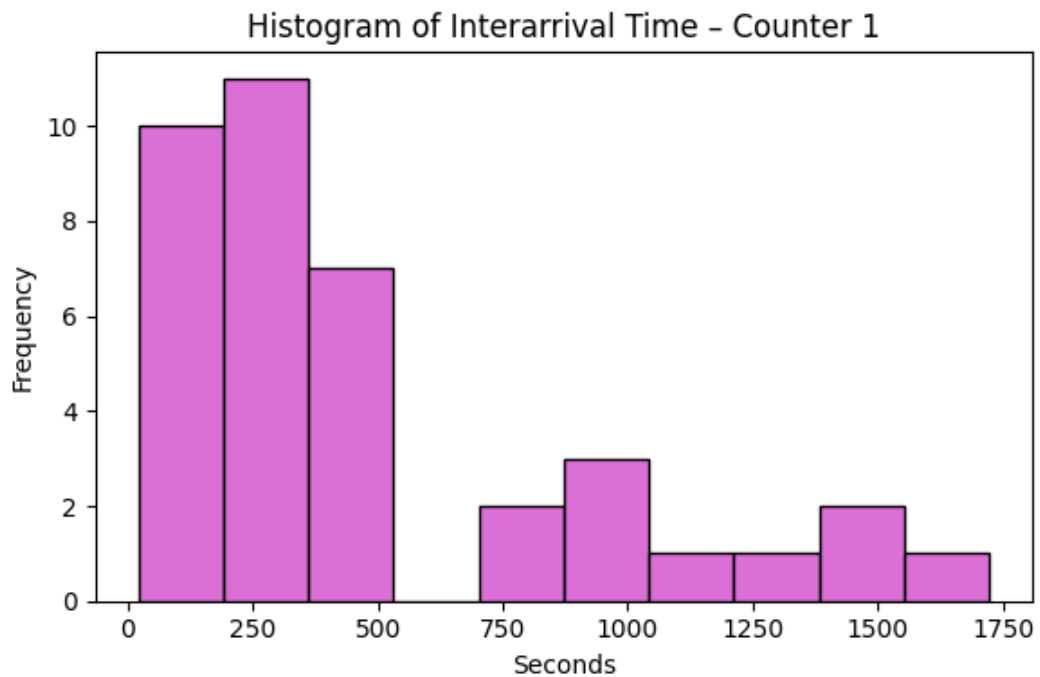


Figure B.1: Histogram of Interarrival Time at Counter 1

The distribution is right-skewed, suggesting exponential characteristics, which aligns with the Poisson arrival assumption of the M/M/1 model.

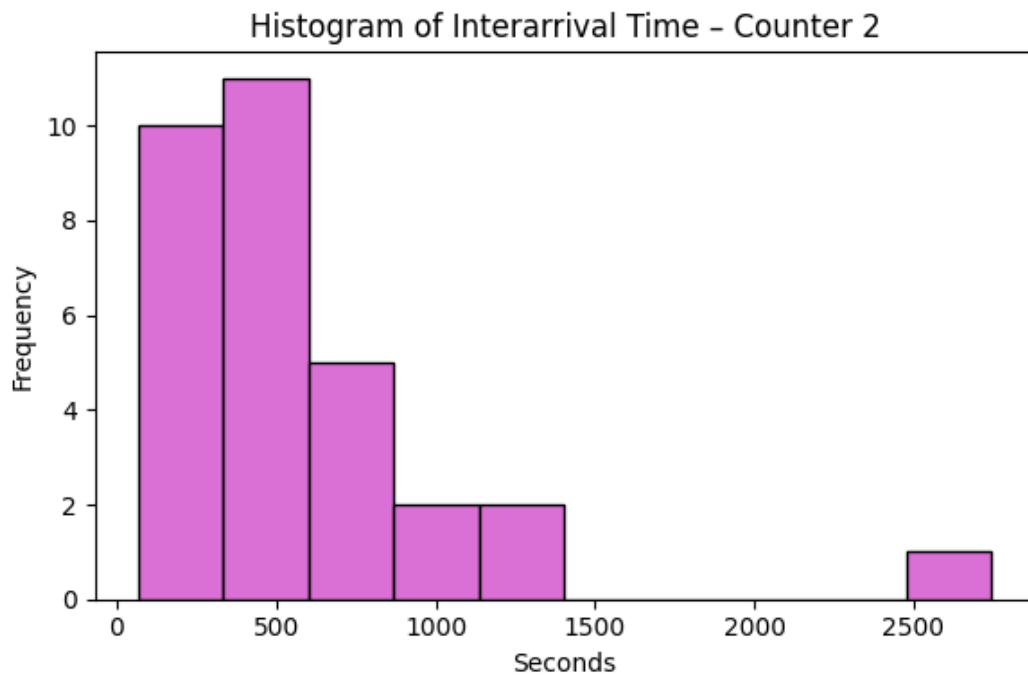


Figure B.2: Histogram of Interarrival Time at Counter 2

The distribution is right-skewed, suggesting exponential characteristics, which aligns with the Poisson arrival assumption of the M/M/1 model.

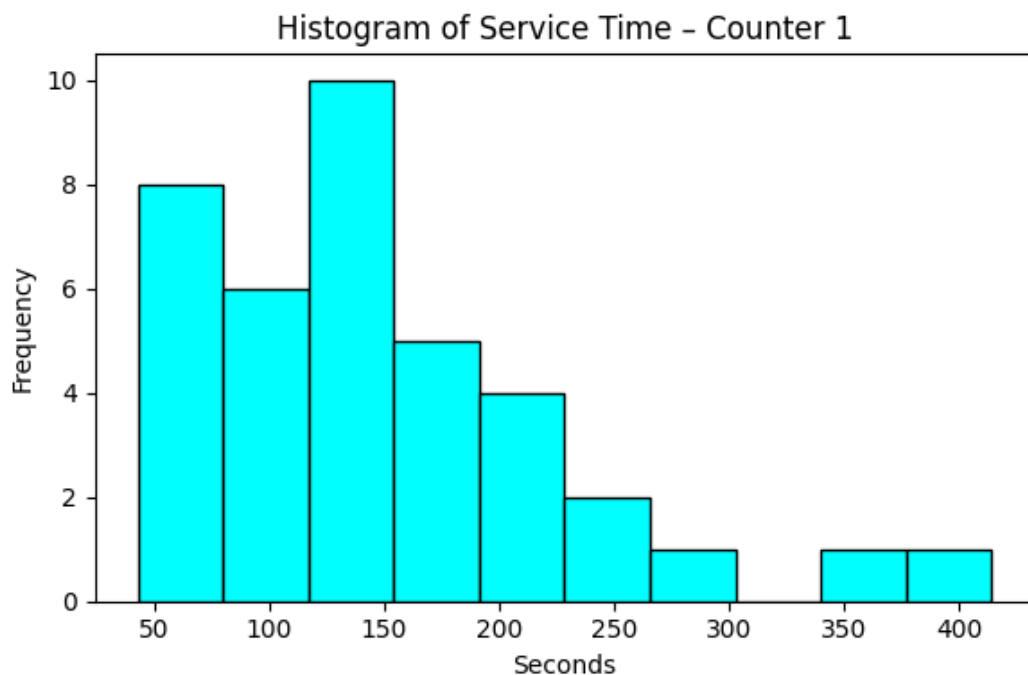


Figure B.3: Histogram of Service Time at Counter 1

Service time shows a similar exponential pattern, with most transactions completed quickly and a few longer interactions. This supports the exponential service time assumption in the M/M/1 model.

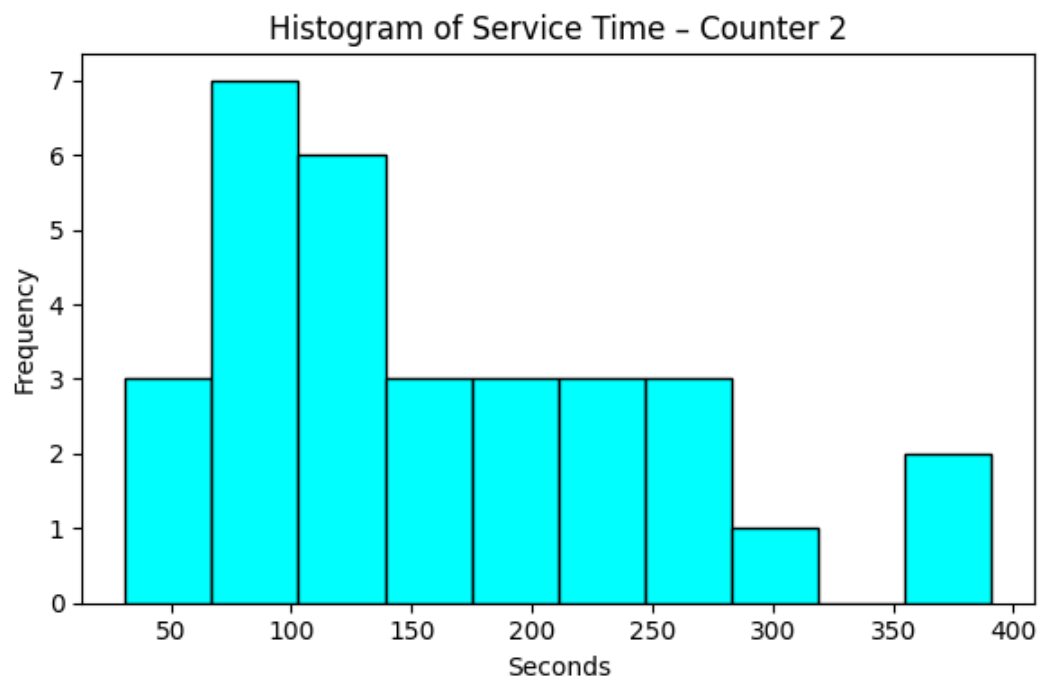


Figure B.4: Histogram of Service Time at Counter 2

Service time shows a similar exponential pattern, with most transactions completed quickly and a few longer interactions. This supports the exponential service time assumption in the M/M/1 model.